Deeper examination of some functions of the `datana` package

Examples

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This document expands on examples of different functions from the datana package (Salas-Eljatib and Campos 2025), implemented in R (R Core Team 2025). The main idea of such a report is to provide most of the utilities and versatility of the aforementioned package's functions.

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1 descstat

1.1 About this function

The function `descstat` is designed to provide a fast, and complete, descriptive statistics table for random variables.

1.2 Example using demographic data

Let's use the following dataframe of the CASEN Chilean survey.

```
library(datana)
df <- datana::casen
df<-subset(df, comuna=="Temuco")
str(df)
```

```
'data.frame': 774 obs. of  21 variables:
 $ id.vivienda      : int  1003001 1016801 1054004 1059302 1059306 1059306 1082806 1082807 1084402 10
 $ id.persona       : int  3 4 3 3 3 4 4 2 5 4 ...
 $ region           : chr  "Región de La Araucanía" "Región de La Araucanía" "Región de La Araucanía"
 $ comuna           : chr  "Temuco" "Temuco" "Temuco" "Temuco" ...
 $ edad             : int  13 7 5 18 15 11 18 15 16 8 ...
 $ sexo             : chr  "1. Hombre" "1. Hombre" "1. Hombre" "1. Hombre" ...
 $ esc              : int  NA NA NA 12 10 NA 11 9 NA NA ...
 $ educ             : chr  "Básica incompleta" "Básica incompleta" "Sin educación formal" "Media huma
 $ personas.hogar   : int  3 4 3 3 4 4 5 2 5 5 ...
 $ tipohogar        : chr  "Nuclear Monoparental" "Extenso Monoparental" "Nuclear Biparental" "Nuclea
 $ activ            : chr  NA NA NA "Inactivos" ...
 $ ytot             : num  0 0 0 11408 0 ...
 $ ytoth            : num  412500 1315652 340000 1324686 1422711 ...
 $ ypch             : num  137500 328913 113333 441562 355678 ...
 $ ytotcor          : num  0 0 0 11408 0 ...
 $ ytotcorh         : num  592500 1315652 340000 1324686 1422711 ...
 $ ypc              : num  197500 328913 113333 441562 355678 ...
 $ mayor.nivel.edu  : chr  "7. Educación Básica" "7. Educación Básica" "4. Prekínder / Kínder (Transi
 $ area.edu.cinef   : chr  NA NA NA NA ...
 $ subarea.edu.cinef: chr  NA NA NA NA ...
 $ previ.salud      : chr  "1. Sistema Público FONASA" "1. Sistema Público FONASA" "1. Sistema Públic
```

1.2.1 For all the numeric variables

```
descstat(df)
```

```
           id.vivienda id.persona    edad     esc personas.hogar        ytot        ytoth        ypch
n              774.000    774.000 774.000 238.000        774.000     774.000      774.000     774.000
Minimum    1003001.000      1.000   0.000   0.000          1.000       0.000        0.000       0.000
Maximum    4585807.000     10.000  65.000  12.000         10.000 1848333.000 10172925.000 2543231.000
Mean       3068761.111      3.381  11.547  10.487          4.099   26162.953  1252683.121  324332.643
Median     3224909.000      3.000  11.000  11.000          4.000       0.000   998901.000  250391.000
Std. Dev.  1114486.564      1.203   6.226   1.818          1.325   91580.991  1002751.638  267611.204
```

```
CV(%)          36.317     35.578  53.917  17.337         32.324     350.041       80.048      82.511
```

Note however that some of the variables, although were recorded as numeric, they are not actual numeric variables (*e.g.,* `id.vivienda` and `id.persona`).

1.2.2 Some of the columns

```
descstat(df,y=c("edad","ytot","ytoth"))
```

```
             edad        ytot        ytoth
n         774.000     774.000      774.000
Minimum     0.000       0.000        0.000
Maximum    65.000 1848333.000 10172925.000
Mean       11.547   26162.953  1252683.121
Median     11.000       0.000   998901.000
Std. Dev.   6.226   91580.991  1002751.638
CV(%)      53.917     350.041       80.048
```

1.2.3 A more detailed output

Let's request a full detailed set of statistics

```
descstat(df,y=c("edad","ytot","ytoth"),full=TRUE)
```

```
                   edad        ytot        ytoth
n               774.000     774.000      774.000
Minimum           0.000       0.000        0.000
Maximum          65.000 1848333.000 10172925.000
Mean             11.547   26162.953  1252683.121
Median           11.000       0.000   998901.000
Std. Dev.         6.226   91580.991  1002751.638
CV(%)            53.917     350.041       80.048
25th Percentile   7.000       0.000   669346.750
75th Percentile  15.000   27736.000  1486498.250
Interq. range     8.000   27736.000   817151.500
Skewness          2.188      12.164        3.212
Kurtosis         12.872     211.169       16.230
```

1.2.4 In Spanish?

You can tailor the output to Spanish by the option `eng`, as follows

```
descstat(df,y=c("edad","ytot","ytoth"),full=TRUE,eng=FALSE)
```

```
                 edad        ytot        ytoth
n             774.000     774.000      774.000
Mínimo          0.000       0.000        0.000
Máximo         65.000 1848333.000 10172925.000
Media          11.547   26162.953  1252683.121
Mediana        11.000       0.000   998901.000
```

```
Desv. St.        6.226   91580.991  1002751.638
CV(%)           53.917     350.041       80.048
Percentil 25     7.000       0.000   669346.750
Percentil 75    15.000   27736.000  1486498.250
Rango Interc.    8.000   27736.000   817151.500
Asimetría        2.188      12.164        3.212
Curtosis        12.872     211.169       16.230
```

1.2.5 Segregated by a factor

The function `descstat` can be applied for the specified variables given in the option `y`, but for each of the levels of a factor variable being specified in the `factvar`, as follows

```
descstat(df,y=c("edad","ytot","ytoth"),factvar="sexo")
```

```
=====================================================================
Factor level: sexo = 1. Hombre
---------------------------------------------------------------------
             edad       ytot        ytoth
n         396.000    396.000      396.000
Minimum     1.000      0.000   132163.000
Maximum    65.000 786667.000 10172925.000
Mean       11.715  25086.197  1230007.899
Median     12.000      0.000   995576.500
Std. Dev.   6.323  74247.087   965320.932
CV(%)      53.976    295.968       78.481

=====================================================================
Factor level: sexo = 2. Mujer
---------------------------------------------------------------------
             edad        ytot       ytoth
n         378.000     378.000     378.000
Minimum     0.000       0.000       0.000
Maximum    53.000 1848333.000 7700000.000
Mean       11.370   27290.984 1276438.116
Median     11.000       0.000 1001041.500
Std. Dev.   6.125  106857.478 1041275.988
CV(%)      53.870     391.549      81.577
```

1.2.6 Segregated by more than one factor

You can expand the option `factvar` to be a vector containing the name of the factors. For instance, if we assume two factors of interest; `sexo` and `educ`, and we want to use the function `descstat` for the same numeric variables as before

```
descstat(df,y=c("edad","ytot","ytoth"),factvar=c("sexo","educ"))
```

```
=====================================================================
Factors level combination: sexo = 1. Hombre & educ = Básica completa
---------------------------------------------------------------------
```

```
            edad       ytot       ytoth
n         24.000     24.000      24.000
Minimum   12.000      0.000  354167.000
Maximum   65.000 786667.000 2615652.000
Mean      15.792  70618.833 1130352.125
Median    14.000      0.000 1011250.000
Std. Dev. 10.509 179039.553  574205.584
CV(%)     66.546    253.529      50.799

========================================================================
Factors level combination: sexo = 2. Mujer & educ = Básica completa
------------------------------------------------------------------------
            edad       ytot       ytoth
n         24.000     24.000      24.000
Minimum   12.000      0.000  301317.000
Maximum   46.000 372000.000 5736250.000
Mean      14.833  26678.083 1198355.292
Median    13.000      0.000  836958.500
Std. Dev.  6.774  75329.390 1124615.185
CV(%)     45.666    282.364      93.847

========================================================================
Factors level combination: sexo = 1. Hombre & educ = Básica incompleta
------------------------------------------------------------------------
             edad       ytot        ytoth
n         164.000    164.000      164.000
Minimum     6.000      0.000   213545.000
Maximum    15.000 150000.000 10172925.000
Mean        9.866  13703.311  1352314.317
Median     10.000      0.000  1090000.000
Std. Dev.   2.272  26756.736  1116265.703
CV(%)      23.024    195.257       82.545

========================================================================
Factors level combination: sexo = 2. Mujer & educ = Básica incompleta
------------------------------------------------------------------------
             edad       ytot       ytoth
n         157.000    157.000     157.000
Minimum     5.000      0.000       0.000
Maximum    15.000 200000.000 7700000.000
Mean        9.522  20720.981 1332898.892
Median     10.000      0.000 1029863.000
Std. Dev.   2.180  36708.092 1071754.305
CV(%)      22.891    177.154      80.408

========================================================================
Factors level combination: sexo = 1. Hombre & educ = Media humanista completa
------------------------------------------------------------------------
            edad       ytot       ytoth
```

```
n         32.000     32.000      32.000
Minimum   17.000      0.000  132163.000
Maximum   40.000 600000.000 2733333.000
Mean      19.312  32994.094  928733.781
Median    18.000      0.000  788699.500
Std. Dev.  4.974 113546.605  576678.532
CV(%)     25.754    344.142      62.093

=====================================================================
Factors level combination: sexo = 2. Mujer & educ = Media humanista completa
---------------------------------------------------------------------
            edad        ytot       ytoth
n         27.000      27.000      27.000
Minimum   16.000       0.000  115226.000
Maximum   53.000 1848333.000 4049976.000
Mean      19.741   88457.444  906632.444
Median    18.000       0.000  781171.000
Std. Dev.  8.065  354699.122  768222.550
CV(%)     40.855     400.983      84.734

=====================================================================
Factors level combination: sexo = 1. Hombre & educ = Media humanista incompleta
---------------------------------------------------------------------
            edad       ytot       ytoth
n         66.000     66.000      66.000
Minimum   14.000      0.000  132496.000
Maximum   40.000 510000.000 5843000.000
Mean      16.879  37216.652 1200682.621
Median    16.000      0.000 1013559.000
Std. Dev.  4.875  92481.396  883121.937
CV(%)     28.885    248.495      73.552

=====================================================================
Factors level combination: sexo = 2. Mujer & educ = Media humanista incompleta
---------------------------------------------------------------------
            edad       ytot       ytoth
n         68.000     68.000      68.000
Minimum   14.000      0.000  132496.000
Maximum   36.000 350000.000 6350000.000
Mean      15.985  28001.441 1264732.750
Median    15.000      0.000  904075.500
Std. Dev.  2.679  59315.652 1027368.571
CV(%)     16.761    211.831      81.232

=====================================================================
Factors level combination: sexo = 1. Hombre & educ = Media técnica profesional completa
---------------------------------------------------------------------
            edad       ytot       ytoth
n          9.000      9.000       9.000
```

```
Minimum   16.000      0.000  712667.000
Maximum   19.000 147221.000 3683834.000
Mean      17.889  29252.556 1439127.556
Median    18.000      0.000 1120276.000
Std. Dev.  0.782  48120.744  964582.024
CV(%)      4.370    164.501      67.025

======================================================================
Factors level combination: sexo = 2. Mujer & educ = Media técnica profesional completa
----------------------------------------------------------------------
            edad      ytot       ytoth
n         10.000    10.000      10.000
Minimum   17.000     0.000  380861.000
Maximum   47.000 33488.000 1408607.000
Mean      21.000  8292.000  778944.700
Median    18.000     0.000  670152.500
Std. Dev.  9.201 12568.495  367440.735
CV(%)     43.816   151.574      47.172

======================================================================
Factors level combination: sexo = 1. Hombre & educ = Media técnica profesional incompleta
----------------------------------------------------------------------
            edad      ytot       ytoth
n         20.000    20.000      20.000
Minimum   14.000     0.000  299754.000
Maximum   18.000 27826.000 1742759.000
Mean      15.750 15375.550  857647.850
Median    16.000 24269.500  807430.500
Std. Dev.  1.209 13431.419  382469.665
CV(%)      7.673    87.356      44.595

======================================================================
Factors level combination: sexo = 2. Mujer & educ = Media técnica profesional incompleta
----------------------------------------------------------------------
            edad      ytot       ytoth
n         17.000    17.000      17.000
Minimum   14.000     0.000  380000.000
Maximum   20.000 57221.000 1850000.000
Mean      16.412 11096.059  985114.882
Median    16.000     0.000  938553.000
Std. Dev.  1.583 17075.201  473890.816
CV(%)      9.648   153.885      48.105

======================================================================
Factors level combination: sexo = 1. Hombre & educ = No sabe
----------------------------------------------------------------------
            edad     ytot       ytoth
n          2.000    2.000       2.000
Minimum   16.000    0.000  544611.000
```

```
Maximum   17.000 8000.000 1386917.000
Mean      16.500 4000.000  965764.000
Median    16.500 4000.000  965764.000
Std. Dev.  0.707 5656.854  595600.284
CV(%)      4.285  141.421      61.671

======================================================================
Factors level combination: sexo = 1. Hombre & educ = Sin educación formal
----------------------------------------------------------------------
            edad       ytot       ytoth
n         79.000     79.000      79.000
Minimum    1.000      0.000  200000.000
Maximum   25.000 322720.000 6447500.000
Mean       5.076  24063.861 1230050.152
Median     5.000      0.000  993351.000
Std. Dev.  4.088  57756.803  992597.589
CV(%)     80.538    240.015      80.696

======================================================================
Factors level combination: sexo = 2. Mujer & educ = Sin educación formal
----------------------------------------------------------------------
            edad       ytot       ytoth
n         75.000     75.000      75.000
Minimum    0.000      0.000  204393.000
Maximum   14.000 450000.000 7700000.000
Mean       4.507  24780.293 1459342.027
Median     5.000      0.000 1264333.000
Std. Dev.  2.069  67260.027 1155707.742
CV(%)     45.908    271.425      79.194
```

1.3 Example using tree-variables data

Let's use the following dataframe of trees from the State of Idaho, in the USA.

```r
library(datana)
df <- datana::idahohd
str(df)
head(df)
```

```
'data.frame': 372 obs. of  5 variables:
 $ plot: int  2 2 3 3 3 4 4 4 5 5 ...
 $ tree: int  1 2 2 5 8 1 2 9 2 4 ...
 $ spp : Factor w/ 10 levels "DF","ES","GF",..: 1 9 3 8 8 8 1 10 1 3 ...
 $ dbh : num  39 48 52 36 38 46 25 29.9 54.9 51.8 ...
 $ toth: num  20.5 33 30 20.7 22.5 18 17 24 29.3 29 ...
plot tree spp dbh toth
1    2    1  DF  39 20.5
2    2    2  WL  48 33.0
3    3    2  GF  52 30.0
```

```
4     3     5  WC   36 20.7
5     3     8  WC   38 22.5
6     4     1  WC   46 18.0
```

1.3.1 For all the numeric variables

```
descstat(df)
```

```
             plot    tree     dbh    toth
n         372.000 372.000 372.000 372.000
Minimum     2.000   1.000  10.000   5.000
Maximum   144.000  10.000 112.000  48.000
Mean       65.280   2.677  36.628  24.313
Median     59.000   2.000  33.750  24.500
Std. Dev.  40.049   1.802  18.159   7.382
CV(%)      61.351  67.320  49.576  30.365
```

Note however that some of the variables, although were recorded as numeric, they are not actual numeric variables (*e.g.,* `plot` and `tree`).

1.3.2 Some of the columns

```
descstat(df,y=c("dbh","toth"))
```

```
               dbh    toth
n         372.000 372.000
Minimum    10.000   5.000
Maximum   112.000  48.000
Mean       36.628  24.313
Median     33.750  24.500
Std. Dev.  18.159   7.382
CV(%)      49.576  30.365
```

1.3.3 A more detailed output

Let's request a full detailed set of statistics

```
descstat(df,y=c("dbh","toth"),full=TRUE)
```

```
                    dbh    toth
n               372.000 372.000
Minimum          10.000   5.000
Maximum         112.000  48.000
Mean             36.628  24.313
Median           33.750  24.500
Std. Dev.        18.159   7.382
CV(%)            49.576  30.365
25th Percentile  22.975  19.000
75th Percentile  46.350  29.500
Interq. range    23.375  10.500
```

```
Skewness          1.014   0.145
Kurtosis          1.383  -0.103
```

1.3.4 In Spanish?

You can tailor the output to Spanish by the option `eng`, as follows

```
descstat(df,y=c("dbh","toth"),full=TRUE,eng=FALSE)
```

```
                  dbh    toth
n             372.000 372.000
Mínimo         10.000   5.000
Máximo        112.000  48.000
Media          36.628  24.313
Mediana        33.750  24.500
Desv. St.      18.159   7.382
CV(%)          49.576  30.365
Percentil 25   22.975  19.000
Percentil 75   46.350  29.500
Rango Interc.  23.375  10.500
Asimetría       1.014   0.145
Curtosis        1.383  -0.103
```

1.3.5 Segregated by a factor

The function `descstat` can be applied for the specified variables given in the option `y`, but for each of the levels of a factor variable being specified in the `factvar`, as follows

```
descstat(df,y=c("dbh","toth"),factvar="spp")
```

```
======================================================================
Factor level: spp = DF
----------------------------------------------------------------------
             dbh   toth
n         55.000 55.000
Minimum   14.000 11.000
Maximum   99.800 42.000
Mean      39.764 25.438
Median    40.000 25.500
Std. Dev. 16.553  6.680
CV(%)     41.630 26.260

======================================================================
Factor level: spp = ES
----------------------------------------------------------------------
             dbh   toth
n          2.000  2.000
Minimum   38.000 24.000
Maximum   55.000 32.000
Mean      46.500 28.000
```

```
Median    46.500 28.000
Std. Dev. 12.021  5.657
CV(%)     25.851 20.203

======================================================================
Factor level: spp = GF
----------------------------------------------------------------------
              dbh    toth
n         111.000 111.000
Minimum    10.200   5.000
Maximum    86.100  47.000
Mean       35.343  24.414
Median     32.200  24.500
Std. Dev.  17.113   7.728
CV(%)      48.419  31.655

======================================================================
Factor level: spp = HW
----------------------------------------------------------------------
             dbh   toth
n          4.000  4.000
Minimum   17.500 18.000
Maximum   31.200 24.000
Mean      21.750 20.250
Median    19.150 19.500
Std. Dev.  6.473  2.630
CV(%)     29.760 12.987

======================================================================
Factor level: spp = LP
----------------------------------------------------------------------
             dbh   toth
n          3.000  3.000
Minimum   19.200 16.000
Maximum   31.000 25.000
Mean      24.733 21.833
Median    24.000 24.500
Std. Dev.  5.934  5.058
CV(%)     23.992 23.166

======================================================================
Factor level: spp = PP
----------------------------------------------------------------------
             dbh   toth
n          2.000  2.000
Minimum   37.100 26.000
Maximum   76.900 40.000
Mean      57.000 33.000
Median    57.000 33.000
```

```
Std. Dev. 28.143  9.899
CV(%)     49.373 29.998

======================================================================
Factor level: spp = SF
----------------------------------------------------------------------
             dbh   toth
n         10.000 10.000
Minimum   10.000  9.000
Maximum   19.100 20.500
Mean      13.550 15.410
Median    12.900 15.650
Std. Dev.  3.159  3.656
CV(%)     23.315 23.728

======================================================================
Factor level: spp = WC
----------------------------------------------------------------------
               dbh    toth
n         136.000 136.000
Minimum    10.100   9.000
Maximum   101.500  40.000
Mean       39.282  23.658
Median     37.750  24.500
Std. Dev.  18.391   6.914
CV(%)      46.819  29.223

======================================================================
Factor level: spp = WL
----------------------------------------------------------------------
             dbh   toth
n         20.000 20.000
Minimum   12.500 14.000
Maximum   70.800 42.500
Mean      31.875 26.350
Median    28.250 27.000
Std. Dev. 13.726  7.681
CV(%)     43.061 29.151

======================================================================
Factor level: spp = WP
----------------------------------------------------------------------
               dbh   toth
n          29.000 29.000
Minimum    11.500 14.000
Maximum   112.000 48.000
Mean       35.590 26.486
Median     28.700 24.500
Std. Dev.  23.170  8.323
```

```
CV(%)       65.103 31.424
```

1.3.6 Segregated by more than one factor

You can expand the option `factvar` to be a vector containing the name of the factors. To explain this example in a better way, let's create a new factor variable, based upon the variable `dbh`, which will be created by using the function `assigncl()`, as follows

```
df <- datana::assigncl(data = df, variable = "dbh")
head(df)
```

```
plot tree spp dbh toth dbh.class
1    2    1  DF  39 20.5         3
2    2    2  WL  48 33.0         4
3    3    2  GF  52 30.0         4
4    3    5  WC  36 20.7         3
5    3    8  WC  38 22.5         3
6    4    1  WC  46 18.0         3
```

Therefore, now we can have two factors of interest; `spp` and `dbh.class`, and we want to use the function `descstat` for the numeric variable `toth`, as follows

```
descstat(df,y="toth",factvar=c("spp","dbh.class"))
```

```
======================================================================
Factors level combination: spp = DF & dbh.class = 1
----------------------------------------------------------------------
             toth
n           8.000
Minimum    11.000
Maximum    22.600
Mean       17.850
Median     18.100
Std. Dev.   3.548
CV(%)      19.877

======================================================================
Factors level combination: spp = GF & dbh.class = 1
----------------------------------------------------------------------
             toth
n          28.000
Minimum     5.000
Maximum    25.000
Mean       15.393
Median     16.000
Std. Dev.   4.843
CV(%)      31.462

======================================================================
```

```
Factors level combination: spp = HW & dbh.class = 1
-----------------------------------------------------------------------
            toth
n          3.000
Minimum   18.000
Maximum   20.000
Mean      19.000
Median    19.000
Std. Dev.  1.000
CV(%)      5.263

=======================================================================
Factors level combination: spp = LP & dbh.class = 1
-----------------------------------------------------------------------
          toth
n            1
Minimum     16
Maximum     16
Mean        16
Median      16
Std. Dev.   NA
CV(%)       NA

=======================================================================
Factors level combination: spp = SF & dbh.class = 1
-----------------------------------------------------------------------
            toth
n         10.000
Minimum    9.000
Maximum   20.500
Mean      15.410
Median    15.650
Std. Dev.  3.656
CV(%)     23.728

=======================================================================
Factors level combination: spp = WC & dbh.class = 1
-----------------------------------------------------------------------
            toth
n         28.000
Minimum    9.000
Maximum   19.000
Mean      13.968
Median    14.000
Std. Dev.  2.986
CV(%)     21.374

=======================================================================
Factors level combination: spp = WL & dbh.class = 1
```

```
------------------------------------------------------------------------
             toth
n           6.000
Minimum    14.000
Maximum    26.000
Mean       19.583
Median     18.500
Std. Dev.   4.152
CV(%)      21.203

========================================================================
Factors level combination: spp = WP & dbh.class = 1
------------------------------------------------------------------------
             toth
n           9.000
Minimum    14.000
Maximum    24.500
Mean       19.667
Median     21.000
Std. Dev.   3.446
CV(%)      17.522

========================================================================
Factors level combination: spp = DF & dbh.class = 2
------------------------------------------------------------------------
             toth
n          14.000
Minimum    14.500
Maximum    31.000
Mean       22.107
Median     21.750
Std. Dev.   4.642
CV(%)      20.996

========================================================================
Factors level combination: spp = GF & dbh.class = 2
------------------------------------------------------------------------
             toth
n          32.000
Minimum    17.000
Maximum    32.000
Mean       23.250
Median     23.000
Std. Dev.   3.853
CV(%)      16.571

========================================================================
Factors level combination: spp = HW & dbh.class = 2
------------------------------------------------------------------------
```

```
          toth
n            1
Minimum     24
Maximum     24
Mean        24
Median      24
Std. Dev.   NA
CV(%)       NA

========================================================================
Factors level combination: spp = LP & dbh.class = 2
------------------------------------------------------------------------
            toth
n          2.000
Minimum   24.500
Maximum   25.000
Mean      24.750
Median    24.750
Std. Dev.  0.354
CV(%)      1.428

========================================================================
Factors level combination: spp = WC & dbh.class = 2
------------------------------------------------------------------------
            toth
n         28.000
Minimum   13.000
Maximum   32.000
Mean      21.618
Median    20.750
Std. Dev.  3.961
CV(%)     18.325

========================================================================
Factors level combination: spp = WL & dbh.class = 2
------------------------------------------------------------------------
            toth
n          8.000
Minimum   15.000
Maximum   30.000
Mean      25.875
Median    28.000
Std. Dev.  5.160
CV(%)     19.942

========================================================================
Factors level combination: spp = WP & dbh.class = 2
------------------------------------------------------------------------
            toth
```

```
n          8.000
Minimum   16.300
Maximum   25.500
Mean      21.788
Median    23.000
Std. Dev.  3.436
CV(%)     15.769

========================================================================
Factors level combination: spp = DF & dbh.class = 3
------------------------------------------------------------------------
            toth
n         16.000
Minimum   17.900
Maximum   34.000
Mean      26.150
Median    25.750
Std. Dev.  5.492
CV(%)     21.004

========================================================================
Factors level combination: spp = ES & dbh.class = 3
------------------------------------------------------------------------
          toth
n            1
Minimum     24
Maximum     24
Mean        24
Median      24
Std. Dev.   NA
CV(%)       NA

========================================================================
Factors level combination: spp = GF & dbh.class = 3
------------------------------------------------------------------------
            toth
n         28.000
Minimum   19.500
Maximum   36.500
Mean      27.925
Median    28.000
Std. Dev.  4.108
CV(%)     14.713

========================================================================
Factors level combination: spp = PP & dbh.class = 3
------------------------------------------------------------------------
          toth
n            1
```

```
Minimum     26
Maximum     26
Mean        26
Median      26
Std. Dev.   NA
CV(%)       NA

======================================================================
Factors level combination: spp = WC & dbh.class = 3
----------------------------------------------------------------------
            toth
n         37.000
Minimum   18.000
Maximum   33.500
Mean      25.481
Median    25.700
Std. Dev.  3.860
CV(%)     15.150

======================================================================
Factors level combination: spp = WL & dbh.class = 3
----------------------------------------------------------------------
            toth
n          3.000
Minimum   29.000
Maximum   37.500
Mean      34.667
Median    37.500
Std. Dev.  4.907
CV(%)     14.156

======================================================================
Factors level combination: spp = WP & dbh.class = 3
----------------------------------------------------------------------
            toth
n          7.000
Minimum   33.300
Maximum   38.500
Mean      35.157
Median    34.000
Std. Dev.  2.011
CV(%)      5.719

======================================================================
Factors level combination: spp = DF & dbh.class = 4
----------------------------------------------------------------------
            toth
n         17.000
Minimum   23.600
```

```
Maximum   42.000
Mean      31.082
Median    30.000
Std. Dev.  5.143
CV(%)     16.548

========================================================================
Factors level combination: spp = ES & dbh.class = 4
------------------------------------------------------------------------
          toth
n            1
Minimum     32
Maximum     32
Mean        32
Median      32
Std. Dev.   NA
CV(%)       NA

========================================================================
Factors level combination: spp = GF & dbh.class = 4
------------------------------------------------------------------------
            toth
n         23.000
Minimum   24.500
Maximum   47.000
Mean      32.743
Median    31.200
Std. Dev.  5.870
CV(%)     17.927

========================================================================
Factors level combination: spp = PP & dbh.class = 4
------------------------------------------------------------------------
          toth
n            1
Minimum     40
Maximum     40
Mean        40
Median      40
Std. Dev.   NA
CV(%)       NA

========================================================================
Factors level combination: spp = WC & dbh.class = 4
------------------------------------------------------------------------
            toth
n         43.000
Minimum   20.500
Maximum   40.000
```

```
Mean      29.728
Median    30.000
Std. Dev.  4.361
CV(%)     14.669

======================================================================
Factors level combination: spp = WL & dbh.class = 4
----------------------------------------------------------------------
            toth
n          3.000
Minimum   23.000
Maximum   42.500
Mean      32.833
Median    33.000
Std. Dev.  9.751
CV(%)     29.699

======================================================================
Factors level combination: spp = WP & dbh.class = 4
----------------------------------------------------------------------
            toth
n          5.000
Minimum   27.200
Maximum   48.000
Mean      34.140
Median    29.000
Std. Dev.  8.992
CV(%)     26.337
```

2 interp

2.1 About this function

The `interp` function provides a way to dealing with interpolation, based in any two series of data of the same length. Cases on which interpolation is needed are usually those dealing with series data, such as: time series of economic, climate and biological varibles.

2.2 Example

In a forestry context, the `interp` function results specially handy for reviewing tree taper data. The idea is that with this function, some important variables (*e.g.,* commercial heights and/or diameters) can be imputed based upon the taper series data. Those variables are key for later on compute commercial volume.

Taper series data must fullfill the following:

1. Decreasing stem diameter along stem-height, *i.e.,* the largest stem-diameter is at the ground level (*i.e.,* stem-height equal to 0) and the smallest-diameter is at the tip of the tree (*i.e.,* stem-height equal to total height).

2. To always exists a section at stump height (generally at 0.3 m), dbh height, and between the mentioned values (generally 0.8m).

3. That the diameter outside bark is always larger than the diameter inside bark for the same height.

4. That the last section (the tip of the tree) has diameter equals 0 (zero) and $hl = h$.

2.2.1 Data

Loading tree taper data:

```
## loading taper data
library(biometrics)
df <- biometrics::tapereuca2
## filter out one tree
df <- subset(df, narb == 1)
head(df[, c("dlcc", "hl")])
```

```
dlcc   hl
1 31.1 0.09
2 28.0 0.80
3 27.6 1.30
4 25.8 2.44
5 24.2 4.88
6 22.6 7.32
```

This tree doesn't have a section at height 3 m, so this is interpolated.

```
library(datana)
## the height will be 3 m
```

```r
new.section <- 3
interpolated <- interp(x = df$dlcc, y = df$hl, ylu = new.section)
interpolated$datares
```

```
x     y interpolated
1  31.10000  0.09        FALSE
2  28.00000  0.80        FALSE
3  27.60000  1.30        FALSE
4  25.80000  2.44        FALSE
5  25.09617  3.00         TRUE
6  24.20000  4.88        FALSE
7  22.60000  7.32        FALSE
8  21.20000  9.76        FALSE
9  19.60000 12.20        FALSE
10 17.90000 14.64        FALSE
11 15.90000 17.08        FALSE
12 13.50000 19.52        FALSE
13  9.80000 21.96        FALSE
14  7.30000 24.40        FALSE
```

The output of the `interp` function is a list of objects

```r
str(interpolated)
```

```
List of 4
 $ intvalues   :'data.frame': 1 obs. of  2 variables:
  ..$ x: num 25.1
  ..$ y: num 3
 $ datares     :'data.frame': 14 obs. of  3 variables:
  ..$ x           : num [1:14] 31.1 28 27.6 25.8 25.1 ...
  ..$ y           : num [1:14] 0.09 0.8 1.3 2.44 3 ...
  ..$ interpolated: logi [1:14] FALSE FALSE FALSE FALSE TRUE FALSE ...
 $ completevals:'data.frame': 1 obs. of  1 variable:
  ..$ xlu.3: num 25.1
 $ datares.full:'data.frame': 14 obs. of  5 variables:
  ..$ x.ori: num [1:14] 31.1 28 27.6 25.8 NA 24.2 22.6 21.2 19.6 17.9 ...
  ..$ y.ori: num [1:14] 0.09 0.8 1.3 2.44 3 ...
  ..$ x    : num [1:14] 31.1 28 27.6 25.8 25.1 ...
  ..$ y    : num [1:14] 0.09 0.8 1.3 2.44 3 ...
  ..$ xlu.3: num [1:14] 25.1 25.1 25.1 25.1 25.1 ...
```

- `intvalues` contains just the values interpolated.
- `datares` contains the resulting series with the interpolated values appended and arranged.
- `completevals` contains the value interpolated for the value given to interpolate.
- `datares.full` contains both the original series and the new series, with the corresponding `completevals` binded.

2.2.2 Failsafe trigger

In cases when the method of interpolation doesn't give desired results, it can be changed. One can specify a different method at the start, or one can specify a failsafe condition to catch on and try another method, *e.g.*:

1. Selecting a method of interpolation at start

```r
## Data that gives negative results on interpolation
hl <- c(0.47, 1.30 , 2.00, 4.70, 8.00, 8.50, 20.00)
dl <- c(82.9, 59.5, 56.4, 49.5, 48.5, 47.0, 0.0)

## interpolation of new hl's given new dl's
newdl <- c(25, 20, 15)
interpolated2 <- interp(x = dl, y = hl, xlu = newdl)
interpolated2$datares
```

```
x         y interpolated
1  15.0 -369.7217         TRUE
2  20.0 -354.5740         TRUE
3  25.0 -296.8190         TRUE
4  82.9    0.4700        FALSE
5  59.5    1.3000        FALSE
6  56.4    2.0000        FALSE
7  49.5    4.7000        FALSE
8  48.5    8.0000        FALSE
9  47.0    8.5000        FALSE
10  0.0   20.0000        FALSE
```

As you can see, the new values are outside the range of original values, moreso, positives and incresing values are needed as a result. Hence, a new method is used when this results occur:

```r
## default method is "spline", now  "linear" is used (see ?datana::interp)
interpolated3 <- interp(x = dl, y = hl, xlu = newdl, method = "linear")
interpolated3$datares
```

```
x        y interpolated
1  82.9  0.47000        FALSE
2  59.5  1.30000        FALSE
3  56.4  2.00000        FALSE
4  49.5  4.70000        FALSE
5  48.5  8.00000        FALSE
6  47.0  8.50000        FALSE
7  25.0 13.88298         TRUE
8  20.0 15.10638         TRUE
9  15.0 16.32979         TRUE
10  0.0 20.00000        FALSE
```

2. Selecting a method of interpolation when a condition is meet The same result can be achieved using the `failsafe` paramater, for this to work a condition to trigger the new failsafe method is needed:

```r
## function to trigger the change of method if the returned
## interpolated values is either less than 0 or larger than the max
## value in the series.
check.value <- function(num, top = stop("give top range")) {
    if (num < 0) {
        return(TRUE)
    } else if (num > max(top)) {
        return(TRUE)
    } else {
        return(FALSE)
    }
}

interpolated4 <- interp(x = dl,
                        y = hl,
                        xlu = newdl,
                        failsafe = TRUE, # enable failsafe catch
                        failsafe.method = "linear", # select a different method to use.
                        failsafe.condition = "|> check.value(top = max(y))")
interpolated4$datares
```

```
x        y interpolated
1  82.9  0.47000        FALSE
2  59.5  1.30000        FALSE
3  56.4  2.00000        FALSE
4  49.5  4.70000        FALSE
5  48.5  8.00000        FALSE
6  47.0  8.50000        FALSE
7  25.0 13.88298         TRUE
8  20.0 15.10638         TRUE
9  15.0 16.32979         TRUE
10  0.0 20.00000        FALSE
```

The `failsafe.condition` is a string with the condition written as if it was concatenated to an "if (value " expression. In the example, a pipe (`|>`) operator is used to pass the interpolated value to the `check.value` function.

3 plotrend

3.1 About this function

The function `plotrend` aims to provide a plot showing the main patter that might be occuring between two variables, expressed as the Y and X axis.

3.2 Example

Let's use the following dataframe

```
library(datana)
df <- datana::maple
head(df)
str(df)
```

```
tree  dbh leaf branch   bole  bark  total
1    1  9.1 0.67  21.30   9.94  1.51  33.42
2    2  9.1 0.54  20.30   6.21  0.63  27.68
3    3  9.9 0.72  19.26  13.93  1.69  35.60
4    4 10.2 0.60  21.77  17.68  1.60  41.65
5    5 16.7 2.90  28.47 101.94 11.37 144.68
6    6 17.2 2.39  26.94  90.26 12.35 131.94
'data.frame': 29 obs. of  7 variables:
 $ tree  : int  1 2 3 4 5 6 7 8 9 10 ...
 $ dbh   : num  9.1 9.1 9.9 10.2 16.7 17.2 17.6 25.1 25.4 25.9 ...
 $ leaf  : num  0.67 0.54 0.72 0.6 2.9 2.39 3.48 7.48 5.83 4.74 ...
 $ branch: num  21.3 20.3 19.3 21.8 28.5 ...
 $ bole  : num  9.94 6.21 13.93 17.68 101.94 ...
 $ bark  : num  1.51 0.63 1.69 1.6 11.37 ...
 $ total : num  33.4 27.7 35.6 41.6 144.7 ...
```

We will fit the following simple regression model

```
m1 <- lm(leaf ~ dbh, data = df)
summary(m1)
```

```
Call:
lm(formula = leaf ~ dbh, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.0875  -1.7636  -0.3398   1.7989  10.7639 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -6.14981    2.39450  -2.568   0.0161 *  
dbh          0.53746    0.07484   7.181 1.01e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 4.537 on 27 degrees of freedom
Multiple R-squared:  0.6564, Adjusted R-squared:  0.6436
F-statistic: 51.57 on 1 and 27 DF,  p-value: 1.008e-07
```

A customary question in regression analysis is to detect any departure from the assumption of the equal variance of the residual. In such a case, the function `plotrend` is suitable, to assess the pattern of the residuals against any predictor variable (Fig. 1) and against the fitted values for the response variable (Fig. 2), too.

```r
datana::plotrend(x = df$dbh, y = residuals(m1),
                 xlab = "Diameter at breast-height (cm)",
                 ylab="Residuals (Kg)")
abline(h = 0)
```

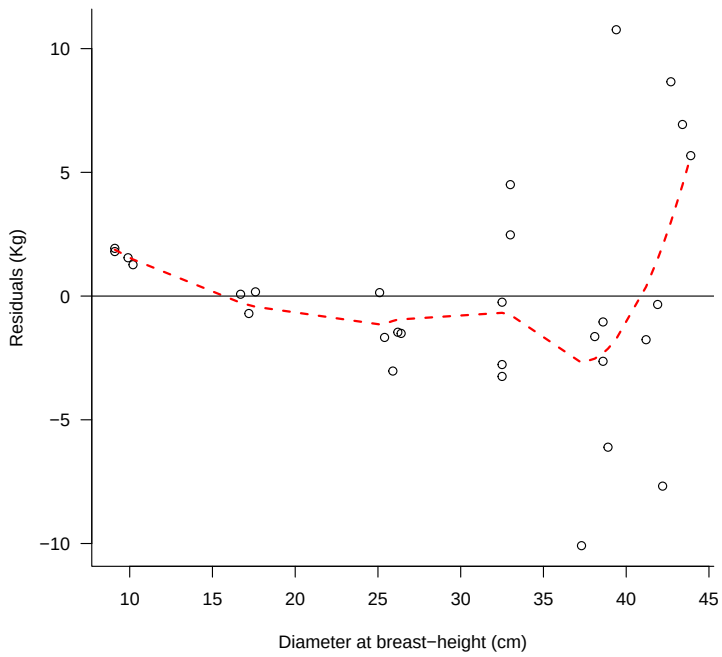

Figure 1: Scatterplot between the residuals of the fitted model and the predictor variable of the model, using the `plotrend` function.

```r
datana::plotrend(x = fitted(m1), y = residuals(m1),
                 xlab = "Fitted values (Kg)",
                 ylab="Residuals (Kg)")
abline(h = 0)
```

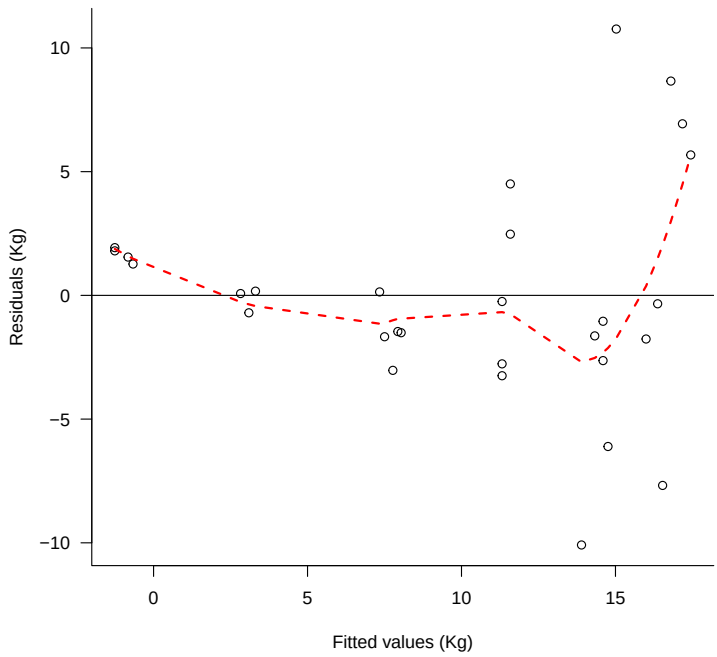

Figure 2: Scatterplot between the residuals of the fitted model and the fitted values for the predictor variable of the model.

References

R Core Team (2025). *R: A language and environment for statistical computing*. `http://www.R-project.org`. R Foundation for Statistical Computing. Vienna, Austria.

Salas-Eljatib, C. and N. Campos (2025). `datana`*: Data and functions to accompany Análisis de datos con R*. R package version 1.1.4. DOI: `10.32614/CRAN.package.datana`. URL: `https://cran.r-project.org/package=datana`.